# Technical Design Document

Technical Design Document | Version 1.0 | 19 October 2026

## 1. Architecture

```
Camera / File     image + EXIF + sensors (gravity, focal length, sensor size)
      |
      v
Pose Detector     BlazePose: 33 landmarks + visibility
      |
      v
Landmark Extractor   heel, foot index, ankle, nose, eyes, ears, mouth,
      |              shoulders, hips; body axis; quality features
      v
Crown Regressor   predicts crown pixel point (BlazePose has no crown landmark)
      |
      v
Geometry Engine   Mode A: proportions model
      |           Mode B: horizon geometry + camera height
      |           Mode C: A4 homography -> camera height -> Mode B
      v
Confidence Calibrator   quality score (0-100) + conformal interval
      |
      v
Result: height_cm, [lo, hi], quality, reason codes
```

## 2. Module contracts

| Module | Input | Output |
|---|---|---|
| Pose Detector | RGB image | 33 landmarks (x, y in px, visibility), or NoPerson / MultiplePeople |
| Landmark Extractor | Landmarks | Named points, body axis vector, head yaw proxy, shoulder and hip tilt, crop margins |
| Crown Regressor | Face landmarks, normalized by a face-scale unit | Crown point (x, y) in px |
| Geometry Engine | Crown, ground point, mode inputs | Height in cm and intermediate values for debugging |
| Confidence Calibrator | Height, quality features, mode | Interval [lo, hi], quality score 0-100, reason codes |

Every module returns a typed result or a typed failure. Failures propagate to the UI as one fixable message.

```
data class Pt(val x: Float, val y: Float)

sealed interface Failure {
    object NoPerson : Failure
    object MultiplePeople : Failure
    object FeetCropped : Failure
    object HeadCropped : Failure
    object HeadTurned : Failure
    object TooDark : Failure
    object PhoneTilted : Failure
    object A4NotFound : Failure
    object UnevenGround : Failure
}

sealed interface Outcome<out T> {
    data class Ok<T>(val value: T) : Outcome<T>
```

```
    data class Fail(val reason: Failure) : Outcome<Nothing>
}
```

# 3. Mathematics

## 3.1 Mode A: anthropometric proportions

All lengths are in pixels along the body's vertical axis.

```
L_body = distance(crown, ground_point)
L_head = distance(crown, chin)
N      = L_body / L_head                  # "heads tall"
r_leg  = distance(hip_mid, ground_point) / L_body
r_sh   = shoulder_width / L_body

height_cm = b0 + b1*N + b2*r_leg + b3*r_sh + b4*sex + b5*age
```

The classical ideal of 7.5 heads is a convention, not a constant, so do not hard-code it. Fit the model instead. Regression shrinks toward the population mean automatically, which is the honest behavior because the image carries no metric scale. The chin is located with a second regressor in the same way as the crown. If the phone is pitched, vertical lengths are foreshortened: rectify using the pitch angle when available, and lower the quality score otherwise.

## 3.2 Mode B: horizon geometry

Known: camera height h_c above the ground, downward pitch θ of the optical axis (from gravity), focal length f in pixels, principal point row c_y, image rows of the feet v_f and the crown v_h. Focal length in pixels is f_px = f_mm x image_width_px / sensor_width_mm, using the width of the active crop. Digital zoom and aspect-ratio crops change it.

```
alpha(v) = atan((v - c_y) / f)          # ray angle below the optical axis
phi(v)   = theta + alpha(v)             # depression angle below horizontal

d = h_c / tan(phi_f)                    # horizontal distance to the person
H = h_c - d * tan(phi_h)
  = h_c * (1 - tan(phi_h) / tan(phi_f))
```

If the crown is above camera level, phi_h is negative and H is greater than h_c, as expected.

**Cross-ratio form.** Take the vertical line through the person in the image. On it, let b = feet, t = crown, h = the point where the horizon line crosses it, and z = the vertical vanishing point (from gravity), as signed coordinates along that line. Cross-ratio is preserved by projection, and in the world these points sit at 0 (feet), H (crown), h_c (camera level) and infinity:

```
(b - h)(t - z)         h_c
--------------  =  - -------
(b - z)(t - h)       H - h_c

H = h_c * (1 - ((b - z)(t - h)) / ((b - h)(t - z)))
```

With no pitch, z goes to infinity and this reduces to H = h_c x (b - t) / (b - h), the person-to-horizon ratio. **Assumptions:** the person is vertical; feet and crown are at about the same distance; roll is corrected using gravity; lens distortion is small or corrected; the ground is flat.

## 3.3 Mode C: A4 homography to camera height

```
1. Detect the A4 quadrilateral; refine corners to subpixel.
2. Fit homography Hm mapping paper coords (X, Y in mm: 0..210, 0..297) to pixels.
3. With intrinsics K: columns h1, h2, h3 of inverse(K) * Hm
     lambda = 1 / norm(h1)
     r1 = lambda*h1,  r2 = lambda*h2,  r3 = r1 x r2,  t = lambda*h3
     choose sign so t_z > 0; orthonormalize R = [r1 r2 r3] with SVD
4. Camera centre in the paper frame: C = -transpose(R) * t
   The paper lies on Z = 0, so h_c = |C_z| (mm).
5. Pass h_c, theta, f and the landmarks into the Mode B formula.
```

The sheet must lie on the same floor plane as the person's feet.

### 3.4 Error floor

In the unit tests, run a Monte Carlo that perturbs pitch by 0.5 degrees, focal length by 2%, landmarks by 3 px and h_c by 1 cm, then records the standard deviation of H. That number is the theoretical error floor for Modes B and C and shows which input deserves the most care.

## 4. Crown regressor

BlazePose has nose, eyes, ears and mouth, but no crown landmark. The crown is estimated from them.

- **Target:** crown offset above the eye line, divided by a face-scale unit (for example ear-to-ear or eye-to-mouth distance), along the body axis.
- **Features:** face landmarks normalized by that scale, plus a head-yaw proxy (nose offset from the ear midpoint).
- **Labels:** you annotate the crown by hand in every training photo. This is a few hundred clicks and the most valuable dataset you will build.
- **Model:** ridge regression first. Report MAE in pixels and in face-scale units, split by subject.
- **Known noise:** hair volume, curls, hats. Use hair type as a stratifying variable in evaluation.

## 5. Confidence calibrator

Two outputs, deliberately separate:

- **Quality score (0-100).** A deterministic score from landmark visibility, margin above and below the body, shoulder and hip levelness, head yaw, phone roll and pitch, and brightness. It rates the photo, not the answer, and the UI says so.
- **Interval.** Split conformal prediction. Collect residuals on a held-out validation set, bucketed by mode and quality bin, and use the 80th percentile of absolute residuals as the half-width. Calibrated by construction if validation photos resemble real-world photos.

## 6. Edge cases

| Case | Detection | Behavior |
|---|---|---|
| Feet cropped out | Heel and foot-index visibility low, or ground point within 2% of the image bottom | Block: "Your feet are cut off. Step back or tilt the phone down." |
| Head turned | Nose-to-ear-midpoint offset above threshold | Warn. Block if severe. Lower quality score. |
| Uneven ground or stairs | Left and right heel heights differ; Mode C sheet and feet not coplanar | Warn. Mode B and C are invalid on sloped ground. |
| Poor lighting | Low mean luminance or low landmark visibility | Ask for more light. Do not process below threshold. |
| Shoes or heels | Not detectable | Ask the user to go barefoot. Otherwise widen the interval. |
| Hair, hats | Not detectable | Widen the interval. Ask to remove hats. |
| Slouching, bent knees | Knee and hip joint angles | Warn. Estimate will read short. |

| Case | Detection | Behavior |
|---|---|---|
| Phone roll | Gravity vector | Correct roll. Block if large. |
| Wide-angle distortion | Camera metadata | Use the main camera. Correct radial distortion if available. |
| Multiple people | More than one pose | Ask to retake or pick one person. |
| A4 not found or partly hidden | Quadrilateral detection fails | Return to Capture with a hint. |

# 7. Performance budgets

- End-to-end processing under 3 seconds on a mid-range phone. Insufficient data to verify the figure; measure on a real older device in phase 7.
- Peak memory low enough for older devices. Downscale the image before pose detection.
- No network permission in the manifest.

# 8. Test plan

- **Synthetic geometry tests.** Build a virtual camera with known h_c, theta and f. Project a vertical segment of known height. The engine must recover H to numerical precision.
- **Golden fixtures.** Python and Kotlin implementations must agree on the same JSON inputs and outputs.
- **Dataset evaluation.** Per-mode, per-phone and per-quality-bin reports using the PRD's validation protocol.
- **Device matrix.** At least one low-end and one mid-range Android 7+ phone for latency and memory.